Final Exam (Assignment) - Productionization of Machine Learning Systems

Start Assignment

- Due Monday by 23:59
- Points 60
- Submitting a file upload
- Attempts 0
- Allowed attempts 1
- Available 29 Jul at 0:00 - 5 Oct at 23:59

Final Exam (Assignment) - Productionization of ML Systems

Total Marks: 60

Q1. Model Deployment (10 Marks)

Discuss different methods of model deployment in brief.

Q2. Model Monitoring and Logging (10 Marks)

Explain the importance of model monitoring and logging.

Q3. FastAPI Deployment with Excel Integration (40 Marks)

Business Context

A bank wants to predict whether a customer is likely to default on a loan. Develop a Random Forest model and deploy it as a REST API using FastAPI. Create an Excel-based interface that enables users to submit customer information and receive the predicted probability of default.

Tasks

1. Load the BankLoan training dataset.
2. Use Defaulter as the dependent (target) variable.
3. Use all variables except SN as independent variables.
4. Develop a Random Forest classification model using 500 trees.
5. Load the BankLoan_Test dataset and estimate the probability of default.
6. Display the predicted probability in the Excel worksheet.
7. Create a FastAPI endpoint (/predict) to return the predicted probability of default.
8. Create a macro-enabled Excel workbook (.xlsm) with a Submit button that calls the FastAPI endpoint and displays the prediction.
9. Submit the Python source code and the Excel workbook with the working Submit button.